

Population Projection Modelling for Fermanagh Community Transport

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1 Introduction

1.1 Research Problem

This project addresses the need for accurate, age-structured population projections for Fermanagh. These help community service providers such as Fermanagh Community Transport (FCT) plan for future service demand. This means that resources can be allocated more efficiently and helps to ensure long term sustainability. Understanding demographic change is particularly important for transport planning as different age groups exhibit varying mobility patterns and accessibility requirements. A lack of projections limits FCT's ability to make decisions about fleet composition, service routing, and infrastructure investments.

1.2 Why it Matters

Fermanagh is experiencing significant population ageing, with rural areas of Northern Ireland to experience a growing proportion of older residents over the coming decades [1]. Loneliness and social isolation are significant issues among the elderly population in rural areas of Northern Ireland, including Fermanagh [2]. Limited transport provision restricts access to social and community activities, increasing the risk of social exclusion and reduced wellbeing among older people [3]. For FCT, this demographic shift means that it is necessary for to transition from standard fixed-route bus services toward more specialised, accessible, and demand-responsive transport options. Older populations generally require door-to-door service with a higher level of accessibility assistance. Younger demographics likely require more frequent services for work and education purposes. Accurate projections enable FCT to:

- Optimise fleet composition between standard and accessible vehicles
- Target driver training programs for specific demographic needs
- Plan service routes based on changing population distributions
- Justify funding applications with robust demographic evidence
- Develop sustainable long-term business strategies

1.3 Approach used

Population projections can be forecasted using different core methodologies:

- **Deterministic Cohort-Component:** Typically used by national statistics agencies [4]. Typically produces a single projection without uncertainty quantification.
- **Trend Extrapolation:** A simple method using past growth trends [5]. Suitable for totals but lacks age-specific detail.
- **Stochastic Simulation:** Models births and deaths as random processes to produce confidence intervals [6].

This project uses a cohort-component framework enhanced with stochastic simulation. Births and deaths are modelled probabilistically using binomial distributions, while migration and ageing are applied deterministically. This hybrid approach provides FCT with probabilistic projections and quantified uncertainty, essential for risk-informed planning.

1.4 Research Question

How will the size and age structure of the Fermanagh population likely change between 2021 and 2031, and what is the statistical uncertainty around these demographic projections?

2 Model Description

2.1 Model Framework and Rationale

A stochastic, discrete-time, age-structured population model was developed. This model is a hybrid stochastic and deterministic because it treats key demographic events such as births and deaths as random processes, while applying migration rates deterministically. Monte Carlo simulation with $M=200$ iterations generates probability distributions for population outcomes, from which 90% confidence intervals are derived, providing FCT with a crucial measure of uncertainty that standard deterministic projections lack.

2.2 Stochastic Framework

Births and deaths are modelled as demographic processes are modelled using binomial distributions. For each process and age group, the annual event count is determined by:

$$\text{Events}_i \sim \text{Binomial}(n = N_i, p = r_i) \quad (1)$$

where i denotes the age group, N_i is the population size, and r_i is the process-specific probability rate. This samples from the binomial PMF:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad (2)$$

2.3 Initial Conditions

The model is initialised with Fermanagh’s 2021 population of 63,585 individuals, distributed across age groups as follows:

Age Group	Population	Proportion
0-14	12,717	20.0%
15-39	18,440	29%
40-64	20,983	33%
65+	11,445	18%
Total	63,585	100.0%

Table 1: Initial population distribution by age group (2021 baseline). These figures are from NISRA [8]

2.4 Model Dynamics

The model progresses annually, updating each group based on three core demographic processes:

- **Fertility:** The model calculates births based on the population of women of childbearing age (15-44). The total number of women in this group is derived from the broader model age groups, with women aged 40-44 constituting 20% of the women in the 40-64 age group.

Fertility Rate Adjustment: Based on Office for National Statistics projections [7], annual fertility rates are adjusted downward by 2% each year from 2022 to 2027. This reflects the projected decline in Northern Ireland’s total fertility rate from 1.81 to 1.63 over this period.

This fertile population is then distributed across six specific age groups using the parameters in table 2.

Age Group	Proportion of Women aged 15-44	Annual Fertility Rate
15-19	0.1677	0.0043
20-24	0.1385	0.0363
25-29	0.1488	0.0950
30-34	0.1688	0.1495
35-39	0.1873	0.0948
40-44	0.1890	0.0205

Table 2: Distribution and fertility rates for women of childbearing age (Sourced from NISRA [9]). The Proportion column shows how the total fertile female population is distributed across age groups. The Fertility Rate gives the annual probability of giving birth for a woman in each group.

The number of births is calculated using equation 2 where N_i is the number of women in age group i and r_i is the corresponding fertility rate. The total number of annual births is the sum across all six age groups.

- **Migration and Ageing:** Each year, a fixed proportion of each group ages out into the next group (e.g., 1/15 of the 0-14 group moves to the 15-39 group annually). Net migration is modelled deterministically by applying migration rates directly to population counts, with migration rates (Table 3) that can be positive (immigration) or negative (emigration).
- **Mortality:** Deaths are subtracted from each group. The number of deaths in each age group is calculated using Equation 2 where N_i is the population of age group i and r_i is the corresponding death rate from Table 3. The total number of annual deaths is the sum across all six age groups.

Age Group	Death Rate	Net Migration Rate
0-14	0.00025	0.005
15-39	0.00070	-0.0125
40-64	0.00421	-0.0075
65+	0.03582	0.008

Table 3: Death and migration rates by age group. Death rates represent annual mortality probabilities. Migration rates show net annual movement, with negative values indicating net emigration from the region. (Data is sourced from NISRA [10], [11])

2.5 Population Update Equations

The population in each age group evolves annually according to the following difference equations:

$$N_{0-14}(t+1) = N_{0-14}(t) + B(t) - A_{0-14}(t) - D_{0-14}(t) + M_{0-14}(t) \quad (3)$$

$$N_{15-39}(t+1) = N_{15-39}(t) + A_{0-14}(t) - A_{15-39}(t) - D_{15-39}(t) + M_{15-39}(t) \quad (4)$$

$$N_{40-64}(t+1) = N_{40-64}(t) + A_{15-39}(t) - A_{40-64}(t) - D_{40-64}(t) + M_{40-64}(t) \quad (5)$$

$$N_{65+}(t+1) = N_{65+}(t) + A_{40-64}(t) - D_{65+}(t) + M_{65+}(t) \quad (6)$$

where:

- $N_i(t)$ = Population in age group i at time t
- $B(t)$ = Total births in year t
- $A_i(t)$ = Number ageing out of group i
- $D_i(t)$ = Deaths in group i
- $M_i(t)$ = Net migration in group i

2.6 Model Assumptions

The model relies on several key assumptions.

- Constant underlying demographic rates (fertility, mortality, migration) throughout the projection period, except for a minor fertility adjustment between 2022-2027.
- The fertility adjustment assumes the projected change in birth rate is linear.
- The female population constitutes exactly 50% of each age group.
- Age distribution within broad groups (e.g., 15-39) remains stable over time
- Proportion of women by age groups (15-44) in Fermanagh is the same as in Northern Ireland and remains consistent throughout the projection period.
- All ageing occurs as discrete annual transitions between groups.
- No international migration effects beyond the specified net migration rates.
- Initial age group proportions are the same in Fermanagh as they are in Fermanagh and Omagh.
- Only women aged between 15 and 44 are fertile.

2.7 Calculating Percentage Population and Errors

Population percentages were calculated for each simulation by dividing age group populations by the total population. For each year and age group, 90% confidence intervals were calculated from the $M = 200$ Monte Carlo simulations as:

$$\text{Confidence Interval} = \text{Mean} \pm \text{Error}$$

where the Error is calculated using:

$$\text{Error} = 1.645 \times \frac{\sigma}{\sqrt{M}}$$

with σ being the standard deviation of the percentages across simulations and 1.645 the z -score for a 90% confidence interval.

2.8 Model Flowchart

The following flowchart illustrates the annual sequence of demographic processes in the population projection model:

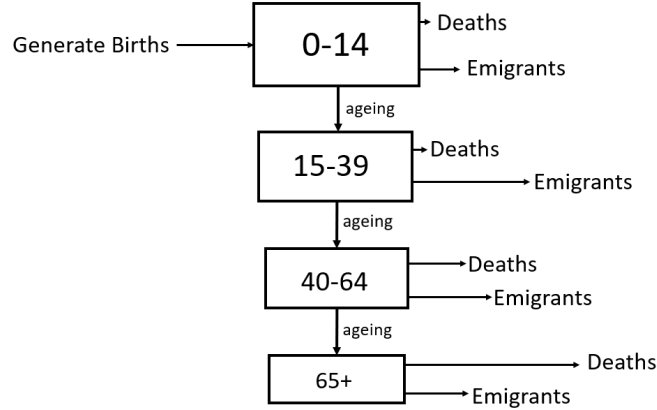


Figure 1: Lifecycle representation of a group of people born in Fermanagh. Births enter the 0-14 age group, then individuals travel downward through age categories via ageing transitions. Each age group experiences deaths and emigration, reducing the amount of people in the next age group. The length of the arrows represents the number of people dying/emigrating with longer representing more people leaving and shorter representing less.

3 Model Results

3.1 Projection Findings

A model was then built to project the evolving age structure. The results, presented as percentages in Figure 2, allow for direct comparison of the demographic composition over time, independent of total population growth. The 90% confidence intervals are included but are not visible on this scale due to the low stochastic variability inherent in projecting large populations with stable demographic rates.

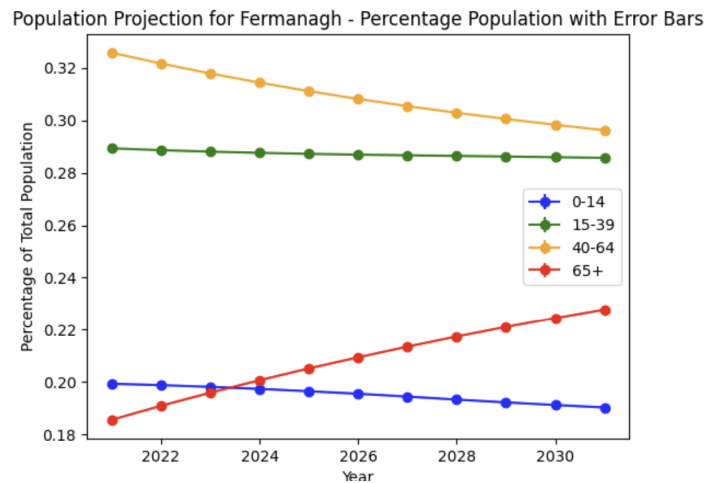


Figure 2: Population percentage projections for Fermanagh (2021–2031) by age group, with 90% Monte Carlo confidence intervals (not visible at this scale).

The graph reveals distinct trends across age groups:

- The youth percentage (0-14) as well as the percentage of those aged 15-39 appears to be declining slightly
- The percentage of the population aged 40-64 drops a larger amount
- The elderly percentage (65+) demonstrates strong growth.

These results reflect lower birth rates and ageing patterns. These projections highlight the current demographic shift toward an older population in Fermanagh. The exact demographic changes for 2031 are detailed in the table below.

Age Group	2021 Population	2031 Projected Population	% Change (2021-2031)	90% CI (2031 Pop.)
0-14	12,717 (20.0%)	12,607 (19.0%)	−0.9%	[12,600, 12,614]
15-39	18,440 (29.0%)	18,907 (28.6%)	+2.5%	[18,904, 18,910]
40-64	20,983 (33.0%)	19,609 (29.6%)	−6.5%	[19,606, 19,612]
65+	11,445 (18.0%)	15,081 (22.8%)	+31.8%	[15,074, 15,088]
Total	63,585 (100.0%)	66,192 (100.0%)	+4.1%	[66,185, 66,199]

Table 4: Population projections for Fermanagh (2021-2031). Values in parentheses show percentage of total population. The percentage change column shows growth/decline in each age group’s absolute size.

The projections in Table 4 reveal a significant demographic shift. The total population is expected to grow by 4.1%. However, this masks largely varying trends within the particular age groups. The most noticeable projection is the 31.8% increase in the population aged 65 and over (an increase of over 3500 people). This contrasts with a 6.5% decline in the 40-64 age group and the relatively maintained population of the younger groups. The confidence intervals, which are extremely narrow, indicate high certainty in the direction and scale of these trends.

3.2 Implications and Recommendations for Fermanagh Community Transport

The projected 31.8% growth in Fermanagh’s population aged 65+ presents a clear strategic plan for the FCT. This should align with the regional Transport Strategy 2035, which prioritises an inclusive and accessibly transport system. This helps to combat social isolation and supports rural communities [13]. Specific actions include:

- **Fleet Accessibility:** Accelerate the change to a fully accessible fleet. This is a direct response to the growing number of mobility needs.
- **Expansion of flexible services:** FCT should use these projections to expand and target its existing demand-responsive transport services, catering for the elderly population.
- **Staff Training:** Implement training in elderly passenger assistance and disability awareness to provide customers with an inclusive experience.
- **Annual review:** Compare these projections with actual service data each year to adjust routes and resources.
- **Funding:** Use these projections to support funding applications, particularly as the department of infrastructure has allocated an additional £1m to community transport providers for this year [12].

3.3 Model Limitations and Extensions

Limitations:

- **Fixed Age Structure:** The model uses four broad age categories (0-14, 15-39, 40-64, 65+) which is consistent with NISRA’s standard demographic reporting. However, these categories mask finer demographic variations that could affect the accuracy of the projections.
- **Constant demographic rates:** Fertility, mortality, and migration rates remain fixed throughout projection period (except minor fertility adjustment). These rates change in reality
- **Equal gender distribution:** The model assumes that there are an equal number of men and women in each age group, which may not be the case. This is especially the case with the 65+ age group, where female longevity typically exceeds male.
- **Deterministic migration:** Since migration is modelled as fixed rates, applied directly to populations rather than as stochastic processes. This approach does not capture the inherent year-to-year variability in migration flows.

- **Stationary population distribution:** The model works under the assumption that proportional distributions within broad age groups remains constant, though internal shifts may occur.

Extensions:

- **Time-varying demographic rates:** Incorporate historical trends to project the change fertility, mortality, and migration rates.
- **Gender-specific modelling:** Separate male and female populations with gender-specific rates for mortality.
- **Stochastic migration:** Implement binomial or Poisson distributions for migration events to better capture variability.
- **Economic and social covariates:** Include local economic indicators, housing development, and healthcare access. As these are drivers of economic change, these could impact fertility, mortality or migration rates.
- **Smaller age groups:** A shift to 5-year age groups (e.g., 0-4, 5-9, ..., 80-84, 75+) would improve the model's utility for FCT's operational planning.

4 Testing the model from 2011-2021

The performance of this model was tested by comparing it to previous data. Everything remained the same except:

- Population was changed to what it was in 2011 (61805)
- Age distribution was changed to what it was in 2011:
 - 20% aged 0-14
 - 33% aged 15-39
 - 32% aged 40-64
 - 14% aged 65+

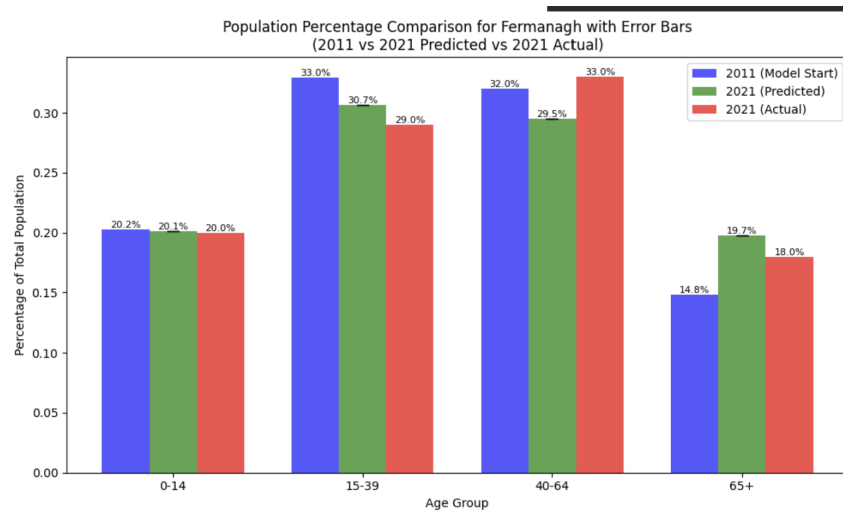


Figure 3: Comparison of actual 2011 population distribution with predicted and actual 2021 distributions.

The model shows reasonable accuracy for younger age groups (0-14 and 15-39) but misjudges the change for older demographics, over-predicting the 65+ population and under-predicting the 40-64 group. Error bars are very minimal for the predictions.

There are problems with using the 2021 model on the 2011 population data:

- **Death Rates:** Using death rates from 2021 likely accounts for improvements in life expectancy, notably for the 65+ group

- **Migration Assumptions:** Migration rates are based on our 2021 assumptions. In reality migration was likely lower in 2011
- **Fertility Rates:** Assumed constant from 2011 (using 2021 rates), but fertility rates likely declined over the decade.

These problems, in conjunction with the limitations of the model (Section 3.3) likely explain the incorrect predictions for the 40-64 and 65+ age groups.

5 Conclusion

This study has projected a significant demographic shift in Fermanagh, notably with the 65+ aged population expected to grow by 31.8% by 2031, an increase from 18% to 24% of the population. There was very little uncertain around with these projections, with 90% confidence intervals for the 2031 population totals spanning less than 0.03% of the mean projection.

For Fermanagh Community Transport, this trend is a clear strategic priority. It implies a large increase in demand for accessible flexible and door-to-door transport services.

The regional Transport Strategy 2035 states that community transport plays a "vital role" in sustaining rural communities and addressing social isolation ([13], pp. 33). As the main provider for Fermanagh, FCT can use the evidence in this report to ensure its service planning and funding applications are directly supported by this regional strategy.

Therefore, the primary recommendation for the FCT is to use these projections to inform fleet composition, routing decisions and funding applications, ensuring its vital services evolve in conjunction with the changing needs of the community.

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